YaleNUSCollege

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Chiang Mai, Thailand

Where is the gas? Exploring non-detections of the CO(1-0) line emission using the Schmidt-Kennicutt Relation at z∼2-3

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NARIT

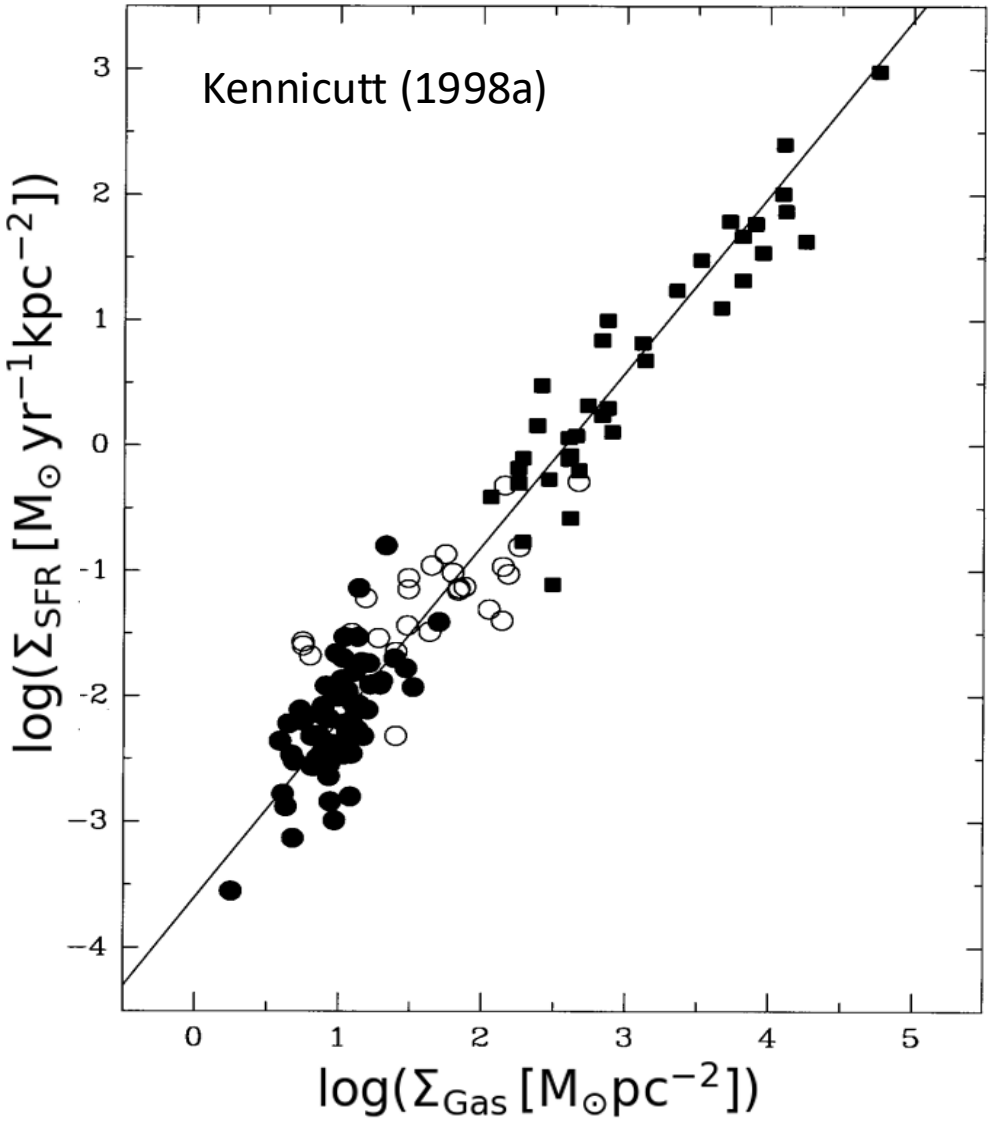


Star Formation in galaxies (in a nutshell)

- Possible to resolve individual molecular clouds in nearby galaxies.
- If not possible, use the Schmidt-Kennicutt (SK) Relation:

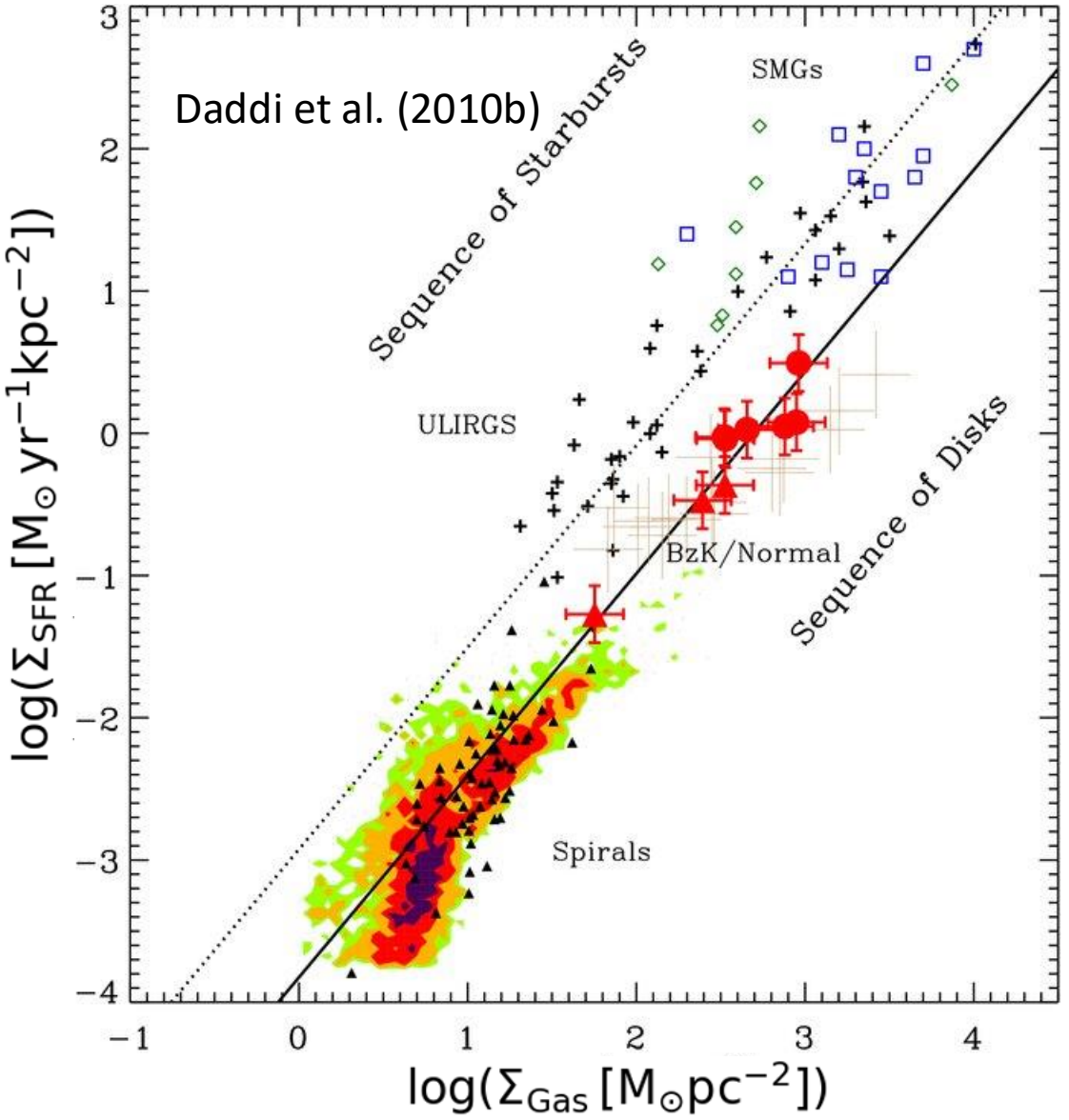
$$\log \Sigma_{SFR} = A + N \log \Sigma_{gas}$$

- SK relation around z ~ 2-3 (cosmic noon) helps us understand star-formation at peak epoch (Förster Schrieber+2020).
- Existing spatially resolved high-z sources atypical so more “normal” star-forming galaxies needed.



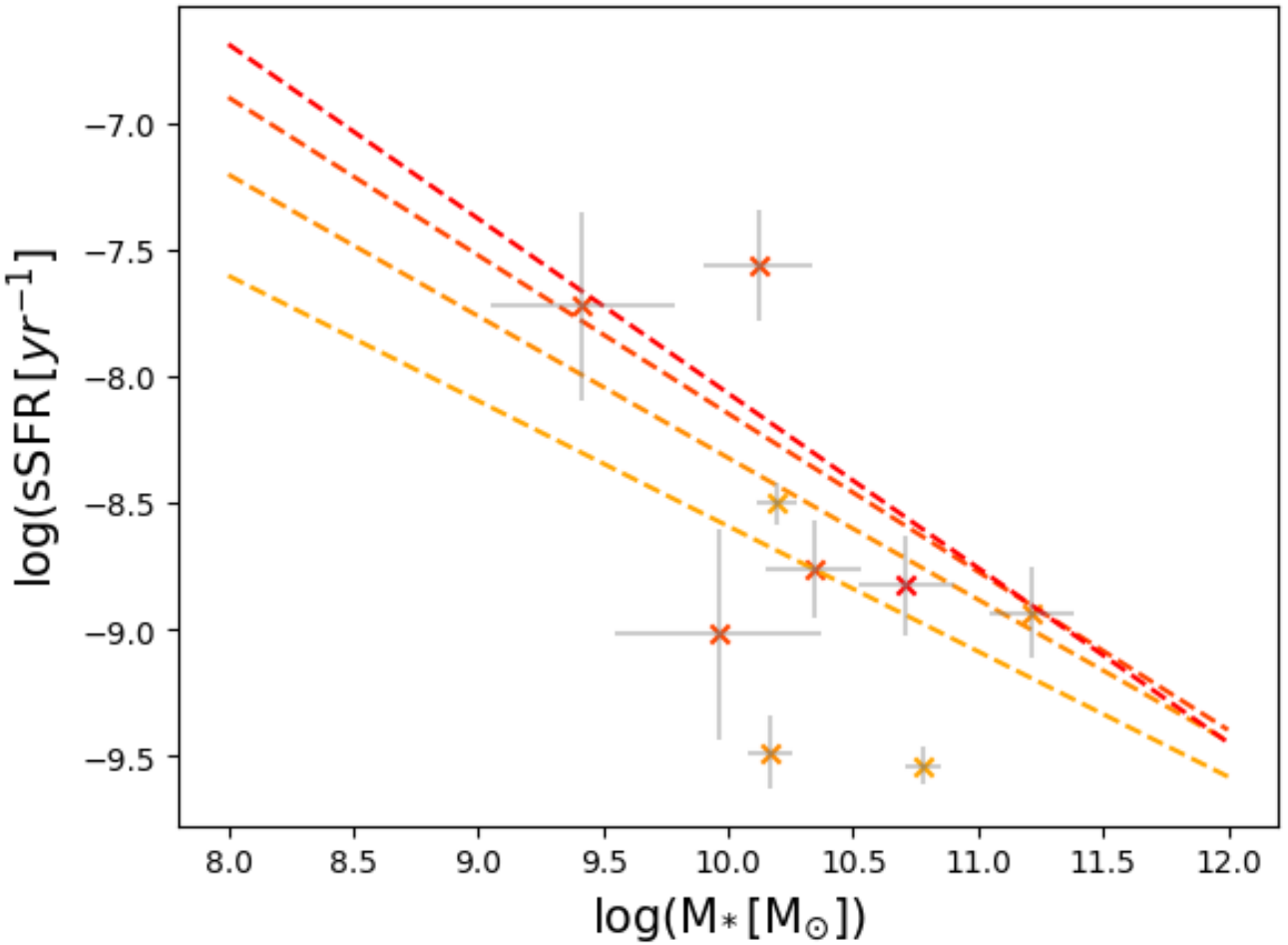
Versions of the SK Law

- **Integrated SK law**
  - Averaged surface density over entire galaxy (Kennicutt 1998a, Wei+ 2010)
  - Luminosity-luminosity plot (e.g. Gao and Solomon 2004 or Dessauges-Zavadsky+2015) at higher-z.
- **Resolved pixel-by-pixel analysis** (e.g. Bigiel+2008, Leroy+2013)
- Daddi+2010b fit two SK laws separately for starbursts and disks/main sequence (MS) each with N = 1.42.



Gas and SFR Tracers for our sample

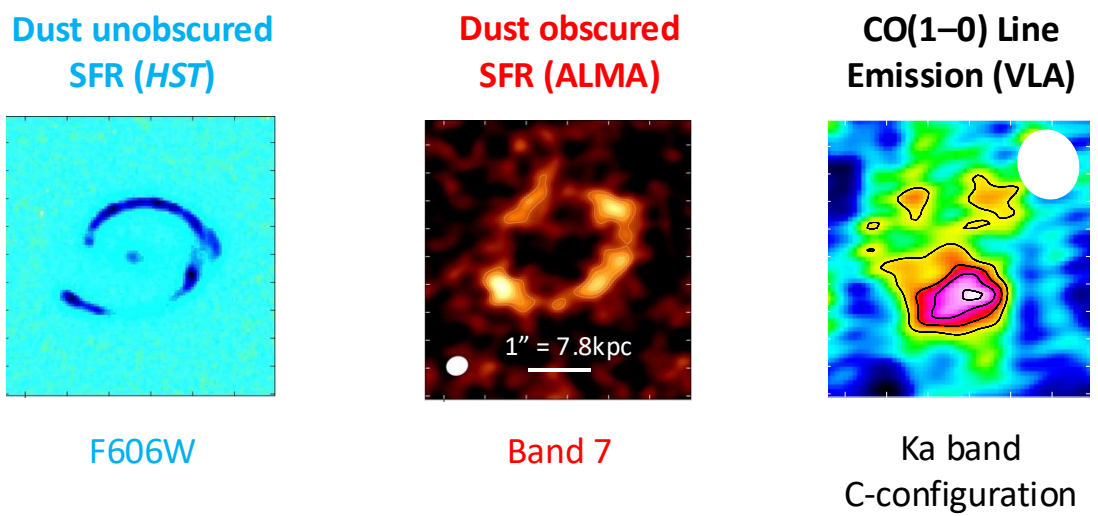
- Gas tracer: CO(1–0) line emission
- Three versions of the CO-H_2 conversion factor (α_{CO}) depending on the goals of analysis:
  - **Disky**: 4.6 (Bolatto+2013)
  - **Starburst/ULIRG/SMG**: 0.8 (Bolatto+2013)
  - **Metallicity+MS offset α_{co}** (Accurso+2017)
- SFR Tracer: *Hubble Space Telescope (HST)* for **rest-frame UV**, and Atacama Large Millimeter Array (ALMA) data for the **IR** (Kennicutt and Evans 2012).



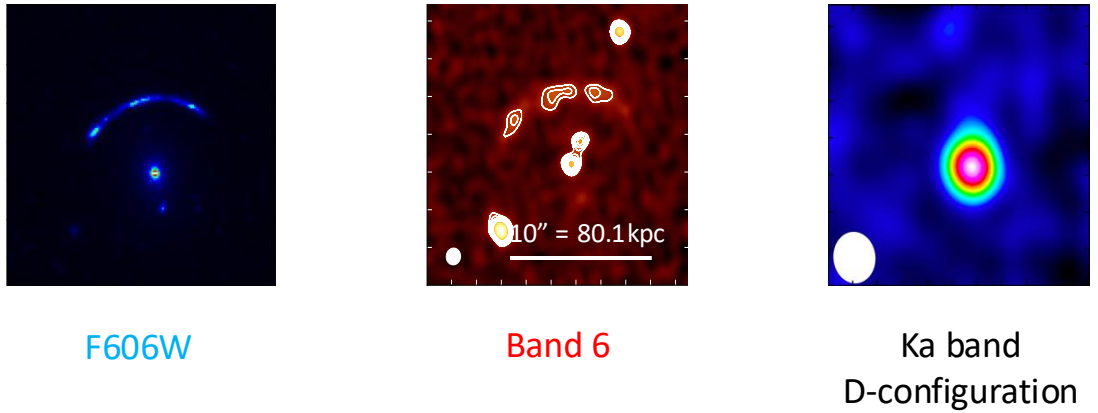
Galaxy Sample

- 9 lensed galaxies at z ∼2–3 from Saintonge+2013, Dessauges-Zavadsky+ 2015.
- Have *HST* and/or ALMA data (SFR) and integrated CO detections (additionally attempted resolved VLA observations of the CO(1–0) line).
- Multiwavelength to fit an SED using Prospector (Johnson+2021 and Leja+2019).
- Historically classified as disks, but Metallicity+MS offset $\alpha_{CO} = 2.67$ to 13.91.

Cosmic Eye (z = 3.07)

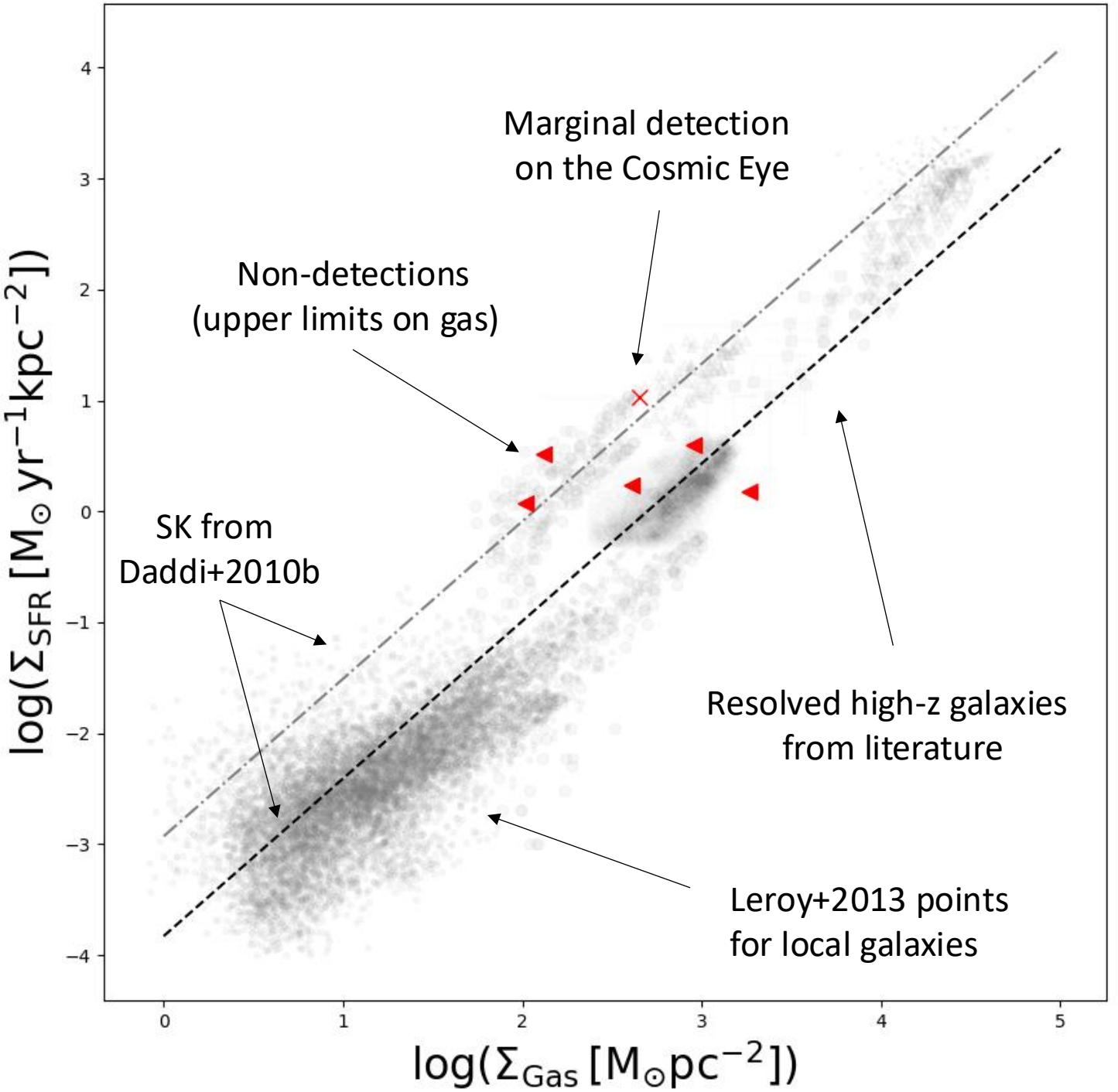


8 o’clock arc (z = 2.73)



Limits on the resolved SK

- Non-detections of CO(1–0) for 5 of the 6 observed galaxies; the Cosmic Eye has a marginal detection.
- Non-detections are not great for any constraints on the SK relation… Therefore, simulations!



Questions

- Can we use our non-detections to understand the SK relation anyway?
  - Constraints on the SK slope
  - Different SFE/depletion timescales (intercepts) for starburst vs disky SK
- Why are we not seeing detections of the CO(1–0) line in our sample of galaxies?
  - α_{CO} conversion factor
  - Velocity gradient

ALMA (Dust obscured)

HST (Dust unobscured)

CO(1–0) Models

SFR (Kennicutt and Evans 2012)

starburst α_{CO}

simobserve on CASA (Common Astronomy Software Applications)

Gas

disky α_{CO}

Simulated VLA observations of CO(1–0)

Starburst SK from Daddi+ 2010b



Disky SK from Daddi+ 2010b



Metallicity+ MS offset α_{CO}

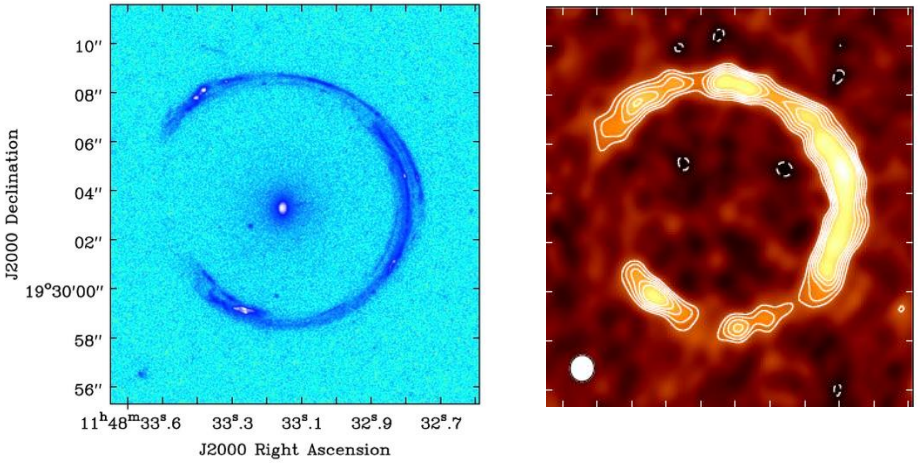
Questions

- Can we use our non-detections to understand the SK relation anyway?
  - Constraints on the SK slope (**no differences seen across slopes = 1, 1.5 and 2**)
  - Different SFE/depletion timescales (intercepts) for starburst vs disky SK
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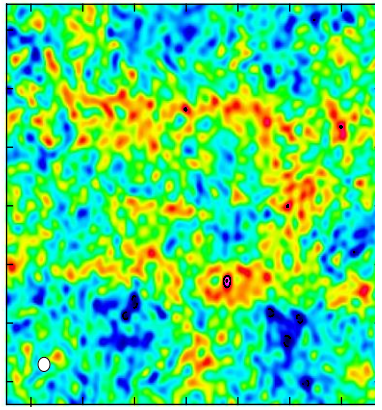
Different intercepts/SFEs on the SK may be due to different α_{CO} conversion factors

- α_{CO} effects on the SK relation also highlighted by e.g. Sandstrom+2013.
- The observed CO(1–0) line is obtained by dividing the gas by the (usual) conversion factor.
- For the Horseshoe, we obtain a 4σ level detection when applying the typical α_{CO} values.

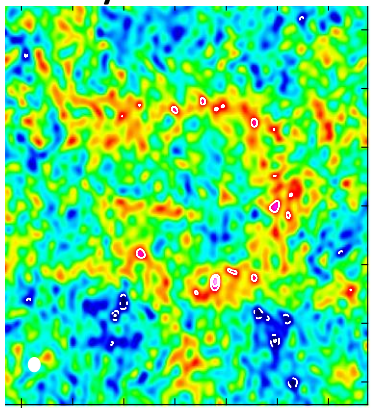
HST and ALMA observations



Starburst-derived CO



Disky derived CO



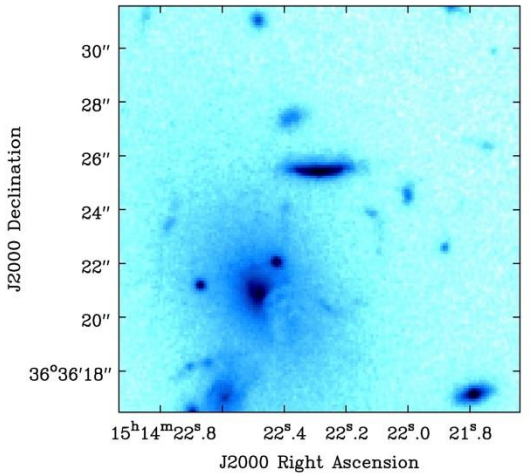
Questions

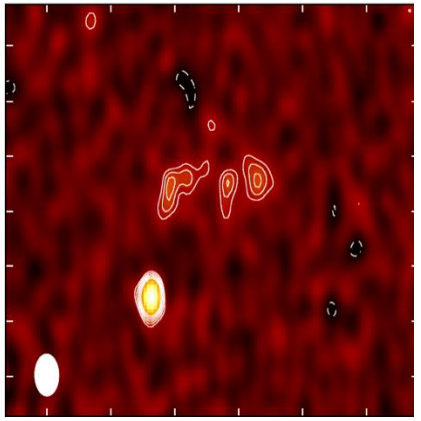
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  - Different SFE/depletion timescales (intercepts) for starburst vs disky SK
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  - Velocity gradient

Accurate α_{CO} estimation seems essential

- For the same Σ_{SFR},for some sources, using the metallicity+MS offset derived α_{CO} allows better match to the non-detection.
- Using the normal disky α_{CO} suggests a spatially resolved detection with existing integration.
- Especially important for known sub-solar metallicity sources.

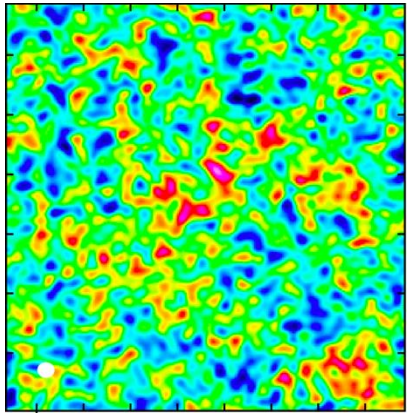
HST and ALMA observations





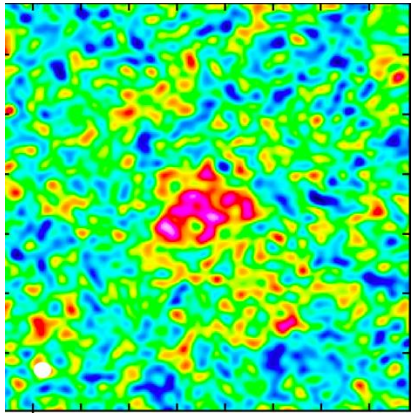
cB58, $\alpha_{CO} = 13.9$

Peak Signal: <3σ



cB58, $\alpha_{CO} = 4.6$

Peak signal: 5σ



Questions

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ALMA (Dust obscured)

HST (Dust unobscured)

CO(1–0) Models

SFR (Kennicutt and Evans 2012)

Gas

Starburst SK from Daddi+ 2010b



Disky SK from Daddi+ 2010b



starburst α_{CO}

disky α_{CO}

Metallicity+ MS offset α_{CO}

simobserve on CASA

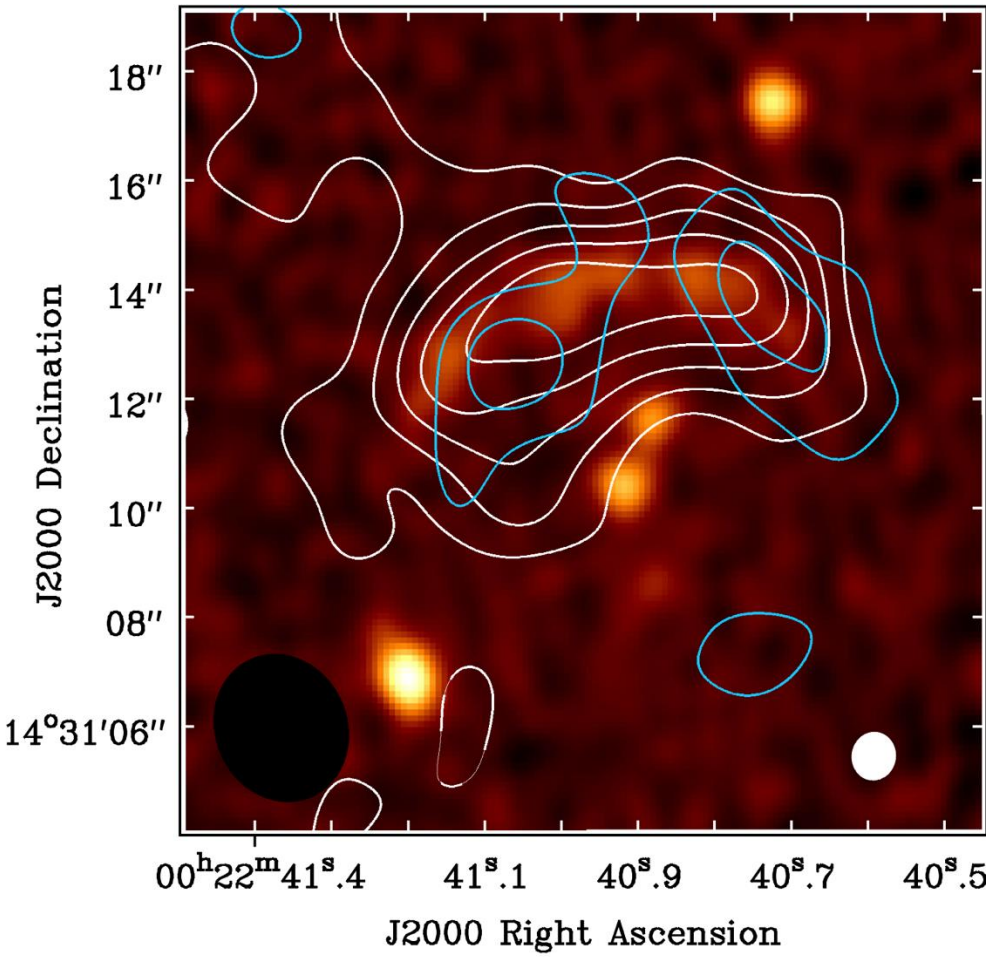
Simulated VLA observations of CO(1–0)

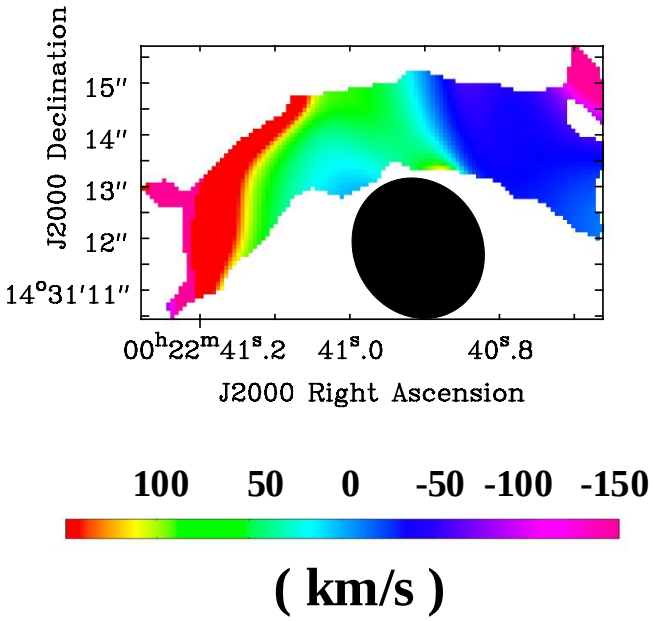
clean model

apply gradient

Velocity gradient on the *clean* model

- We can modify the *clean* model to incorporate gas kinematics.
- Modelling gas kinematics introduces further constraints on signal.
- CO(1–0) emission **without** any kinematics across the source in white, CO(1–0) emission with the gradient (far right) applied to the *clean* model in blue.





Conclusions

- We had 5 non-detections in the CO(1–0) for 6 of the sample of galaxies that were observed.
- Where's the gas? Its probably there, just hidden away due to unfavorable conversion factors and velocities:
  - Proper metallicity derived α_{CO} estimates to be used
  - Kinematics of the gas need to be considered as an independent factor
- The tools developed in this analysis will help us choose suitable targets for follow-up observations.
  - The Cosmic Horseshoe, one of the galaxies in the sample, is perhaps **not** a good candidate for future resolved studies, given an $\alpha_{CO} \sim 15$.

Extra Slides

Full Sample

Source Name	α	δ	z_{spec}
Eye	$21^h35^m12.7^s$	$-01^d01^m42.9^s$	3.074
Horseshoe	$11^h48^m33.1s$	$+19^d30^m03.2^s$	2.379
8:00arc	$00^h22^m40.9s$	$+14^d31^m14.0^s$	2.730
cB58	$15^h14^m22.2s$	$+36^d36^m25.2^s$	2.729
J0744	$07^h44^m47.8s$	$+39^d27^m25.7^s$	2.209
J0712	$07^h12^m17.5s$	$+59^d32^m16.3^s$	2.646
A68-h7	$00^h37^m01.4s$	$+09^d10^m22.3^s$	2.15
A68-C0	$00^h37^m07.4s$	$+09^d09^m26.6^s$	1.5864
A68-HLS115	$00^h37^m09.5s$	$+09^d09^m03.8^s$	1.5869

ALMA Observations

Source	ν (GHz)	S mJy	σ_{rms} (mJy/beam)	Beam arcsec(″)×arcsec(″)	ID	PI
Eye	**340**	2.77 ± 0.35	0.035	0.40×0.34	2013.1.00139.S	N Tohru
	465	$< 3\sigma$	7	0.33×0.31	2013.1.00139.S	N Tohru
8:00arc	**230**	1.86 ± 0.23	0.03	0.89×0.83	2015.1.01157.S	S Malhotra
cB58[a]	**230**	0.56 ± 0.12	0.027	1.53×0.90	2013.1.00262.S	S Malhotra
					2015.1.01157.S	S Malhotra
	340	1.39 ± 0.29	0.075	1.61×0.91	2019.1.01148.S	C Sharon
Horseshoe	**340**	2.76 ± 0.31	0.016	1.07×0.98	2019.1.01148.S	C Sharon
	230	$< 3\sigma$	0.085	1.44×1.01	2013.1.00262.S	S Malhotra

Existing low-resolution CO detections

Source Name	CO line	$S\Delta\nu$ (Jy km/s)	Reference
Eye	CO(1–0)	0.077 ± 0.013	Riechers et al. (2010)
	CO(3–2)	0.50 ± 0.07	Coppin et al. (2007)
Horseshoe	CO(3–2)	0.44 ± 0.18	Saintonge et al. (2013)
8:00arc	CO(3–2)	0.85 ± 0.24	Saintonge et al. (2013)
cB58	CO(1–0)	0.052 ± 0.013	Riechers et al. (2010)
	CO(3–2)	0.37 ± 0.08	Saintonge et al. (2013)
A68-h7	CO(3–2)	1.12 ± 0.22	Dessauges-Zavadsky et al. (2015)
A68-C0	CO(2–1)	1.89 ± 0.28	Dessauges-Zavadsky et al. (2015)
A68-HLS115	CO(2–1)	2.00 ± 0.30	Dessauges-Zavadsky et al. (2015)

Observations

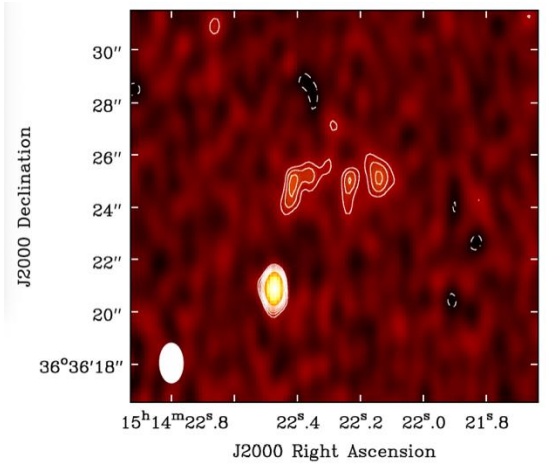
Table 4. Summary of new VLA CO(1–0) observations

Source	Array	$\nu_{\rm obs}$ (GHz)	$S^{\rm a}$ (mJy)	σ_{rms} (mJy/beam)	Time (hr)	Beam arcsec(″)×arcsec(″)	ID
Eye	C	28.2944	0.245 ± 0.044	0.012	29.7	1.26 ×1.09	14B-278, 16A-077
8:00arc	D	30.4146	$< 3\sigma$	0.04	5.22	2.76 ×2.43	18A-114
cB58	C	30.9121	$< 3\sigma$	0.029	14.08	0.91 ×0.74	14B-278, 16A-077
Horseshoe	...	33.1140	...	...	...		
J0744	D	35.9212	$< 3\sigma$	0.05	2.11	2.14 ×1.54	15B-208
J0712	D	31.6158	$< 3\sigma$	0.03	3.14	2.68 ×2.35	15B-208
A68-h7	D	36.5940	$< 3\sigma$	0.05	1.30	2.75 ×2.16	18A-114
A68-C0	...	44.5854	...	...	...		
A68-HLS115	...	44.5768	...	...	...		

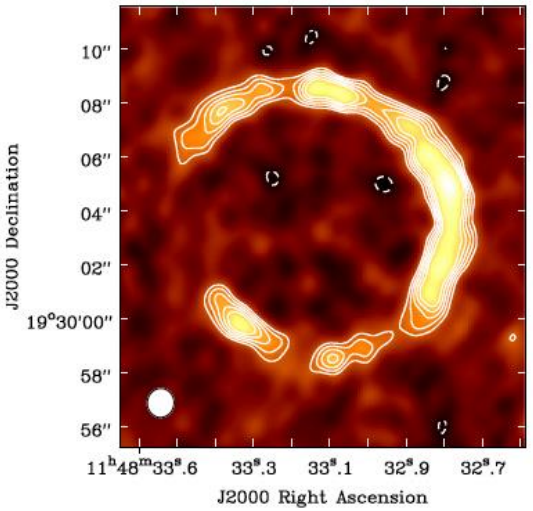
[a]Only the Eye has a reliable detection.

ALMA Maps

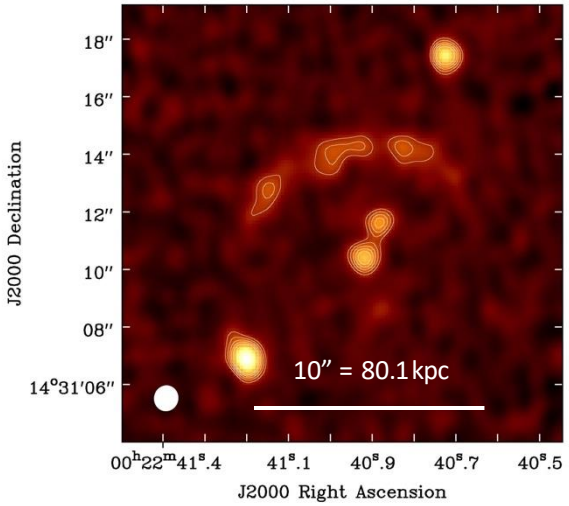
cB58



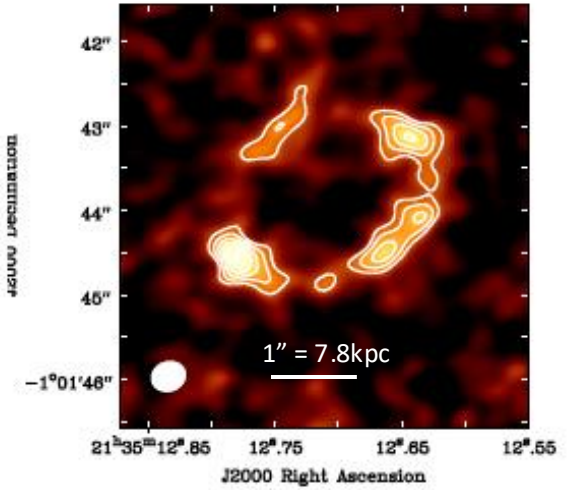
Horseshoe

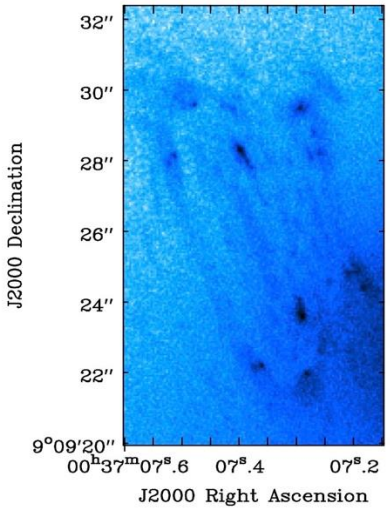


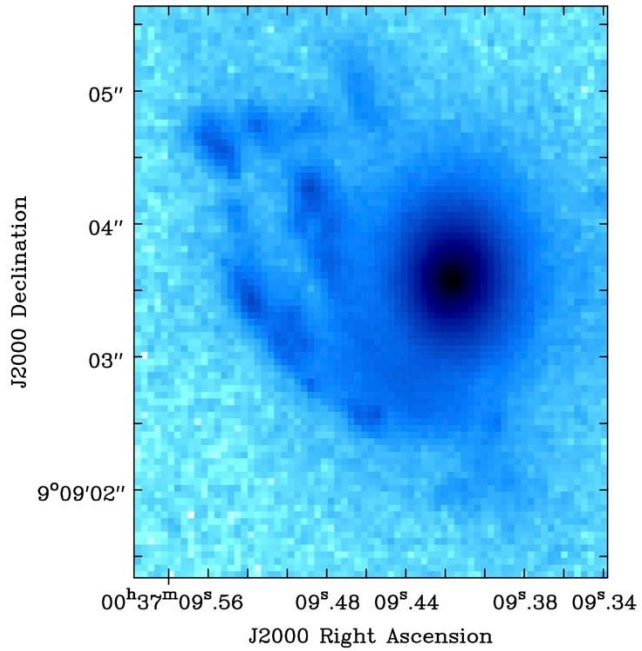
8:00 arc

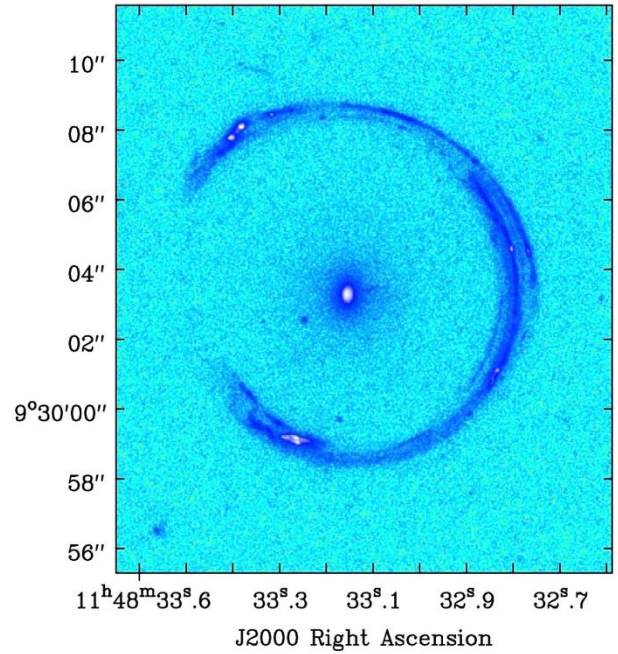


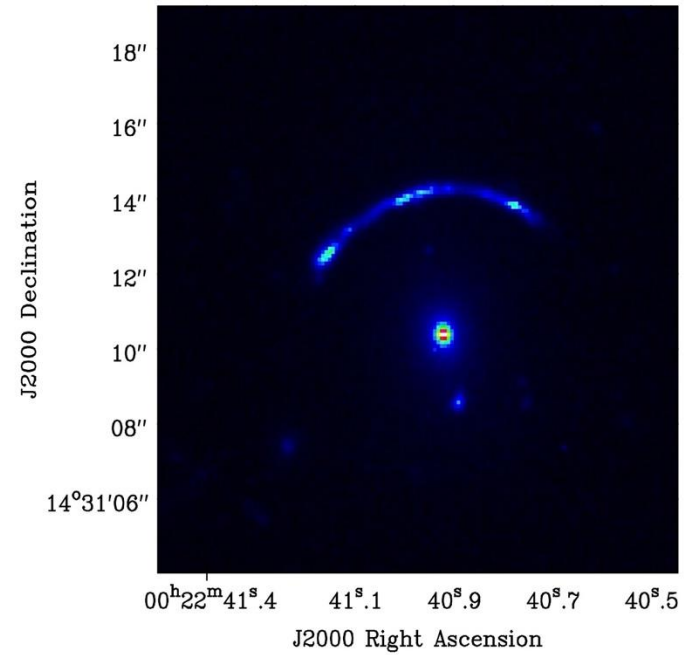
Eye

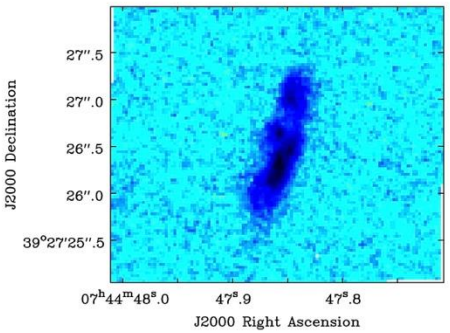


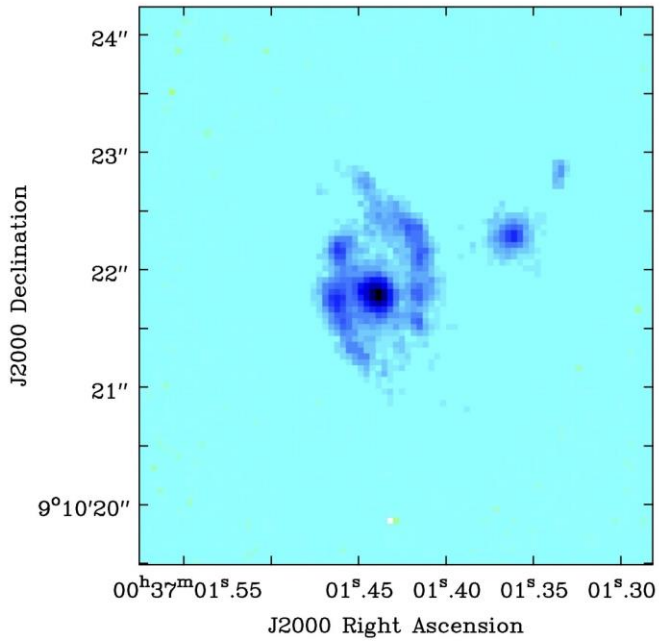

J2000 Declination
32″
30″
28″
26″
24″
22″
9°09′20″
00h37m07s.6
07s.4
07s.2
J2000 Right Ascension

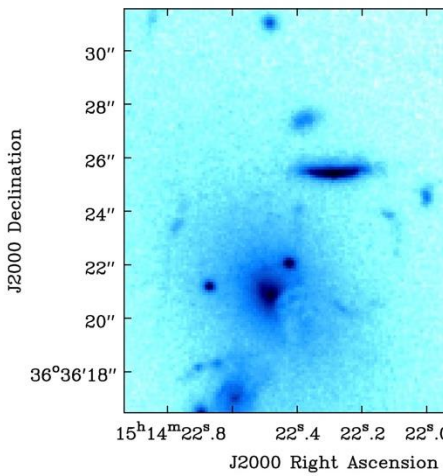

J2000 Declination
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04″
03″
9°09′02″
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09s.48
09s.44
09s.38
09s.34
J2000 Right Ascension

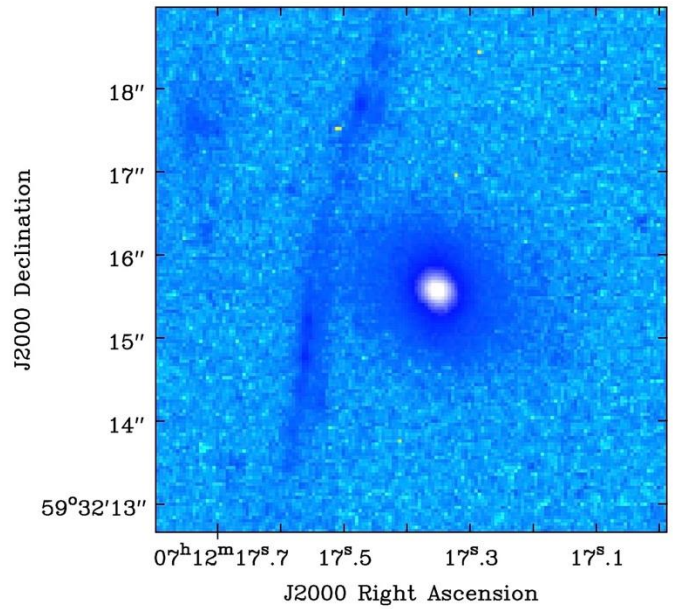

10″
08″
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32s.7
J2000 Right Ascension

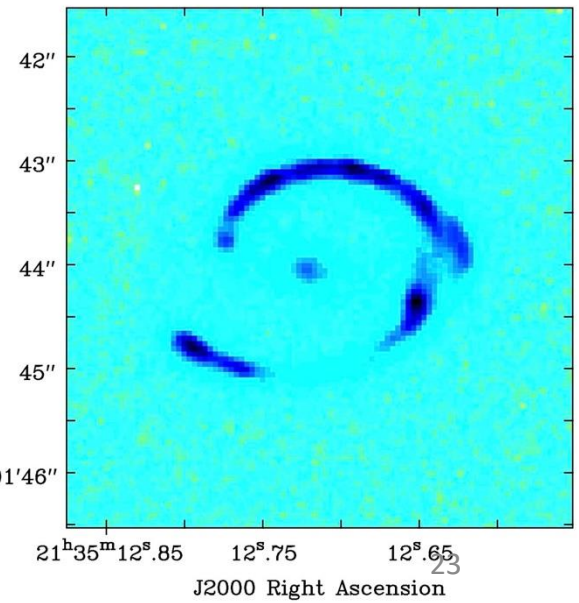

J2000 Declination
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41s.1
40s.9
40s.7
40s.5
J2000 Right Ascension


J2000 Declination
27″.5
27″.0
26″.5
26″.0
39°27′25″.5
07h44m48s.0
47s.9
47s.8
J2000 Right Ascension


J2000 Declination
24″
23″
22″
21″
9°10′20″
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01s.45
01s.40
01s.35
01s.30
J2000 Right Ascension


J2000 Declination
30″
28″
26″
24″
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20″
36°36′18″
15h14m22s.8
22s.4
22s.2
J2000 Right Ascension


J2000 Declination
18″
17″
16″
15″
14″
59°32′13″
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17s.5
17s.3
17s.1
J2000 Right Ascension


42″
43″
44″
45″
01′46″
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12s.75
12s.65
J2000 Right Ascension

Prospector Output Sample

